B. SC. ENGG. IN MSE, RU, CURRICULUM 2024-2025

Summary of Courses

First Year First Semester Examination, 2025

Course Code	Course Title	Marks	Credits	Contact Hours/Week
MSE1111	Introduction to Materials Science and Engineering	50	2	2
MSE1121	Crystallography-1	50	2	2
MSE1122	Crystallography-I Sessional	25	1	3
MATH1111	Matrices, Vector Analysis and Co-ordinate Geometry	75	3	3
PHY1111	Electricity and Magnetism	75	3	3
PHY1112	Electricity and Magnetism Sessional	25	I	3
CHEM1111	Organic Chemistry	50	2	2
CHEM1112	Organic Chemistry Sessional	25	1	3
ENG1111	Technical and Communicative English	50	2	2
MSE1112	Engineering Drawing	25	1	3
	Total	**450**	**18**	**26**

First Year Second Semester Examination, 2025

Course Code	Course Title	Marks	Credits	Contact Hours/Week
MSE1211	Crystallography-II	50	2	2
MSE1212	Crystallography-II Sessional	25	1	3
MSE1221	Solid and Fluid Mechanics	75	3	3
MSE1222	Solid and Fluid Mechanics Sessional	25	1	3
MATH1211	Calculus and Differential Equations	75	3	3
CHEM1211	Physical and Inorganic Chemistry	75	3	3
CHEM1212	Physical and Inorganic Chemistry Sessional	25	1	3
ECON1211	Industrial Economics	50	2	2
ME1212	Mechanical Engineering Workshop	25	1	3
MSE1210	Viva-Voce	50	2	
	Total	**475**	**19**	**25**

Second Year First Semester Examination, 2026

Course Code	Course Title	Marks	Credits	Contact Hours/Week
MSE2111	Materials Thermodynamics	75	3	3
MSE2112	Materials Thermodynamics Sessional	25	1	3
MSE2121	Electronic Properties of Materials	75	3	3
MSE2122	Electronic Properties of Materials Sessional	25	1	3
EEE2111	Electrical Engineering	75	3	3
EEE2112	Electrical Engineering Sessional	25	1	3
MATH2111	Numerical, Statistical and Advanced Mathematical Methods	75	3	3
PHY2111	Waves, Optics and Optical Properties	75	3	3
PHY2112	Waves, Optics and Optical Properties Sessional	25	1	3
ACCO2111	Industrial Management and Accountancy	50	2	2
	Total	**525**	**21**	**29**

Fourth Year First Semester Examination, 2028

Course Code	Course Title	Marks	Credits	Contact Hours/ Week
MSE4111	Materials Characterization-I	75	3	3
MSE4112	Materials Characterization-1 Sessional	25	1	3
MSE4121	Plastic and Rubber Technology	75	3	3
MSE4122	Plastic and Rubber Technology Sessional	25	1	3
MSE4131	Magnetic and Dielectric Materials	75	3	3
MSE4132	Magnetic and Dielectric Materials Sessional	25	1	3
MSE4141	Waste Management, Industrial Safety and Environmental Issues	50	2	2
MSE4151	Surface Science and Engineering	50	2	2
MSE4152	Surface Science and Engineering Sessional	25	1	3
MSE4162	Industrial Training/Study tour/Field work	50	2	
MSE4172	Research Project-I	25	1	
	Total	**475**	**20**	**25**

Fourth Year Second Semester Examination, 2028

Course Code	Course Title	Marks	Credits	Contact Hours/Week
MSE4211	Nano-and Biomaterials	75	3	3
MSE4212	Nano-and Biomaterials Sessional	25	1	3
MSE4221	Materials Characterization-II	75	3	3
MSE4231	Computational Materials Science	75	3	3
MSE4232	Computational Materials Science Sessional	25	1	3
MSE4241	Engineering Materials	50	2	2
MSE4251	Fiber and Textile Engineering	75	3	3
MSE4252	Fiber and Textile Engineering Sessional	25	1	3
MSE4262	Research Project-II	50	2	
MSE4210	Viva-Voce	50	2	
	Total	**550**	**21**	**23**

Course Outline: First Year First Semester Examination, 2025

Course Code	Course Title	Credits	Marks	Contact Hours/Week
MSE1111	Introduction to Materials Science and Engineering	2	50	2

Course Types: Theoretical [x] Sessional/Practical [ ] Project/Training [ ] Viva-voce [ ]

Teaching-learning strategies: Lectures/Exercises/Quizzes/Class Tests/Attendance

Assessment Strategies: Attendance (10%); Quizzes/Class Test (20%); Examination (70%)

Course Description: This course is designed to familiarize students with the fundamental concepts of materials science and engineering, which will serve as a foundation for understanding specialized courses in the field of Materials Science and Engineering. Thus, this course introduces the type of materials, structure, properties, characteristics, and applications with special emphasis on the relationships between internal structure and properties.

Intended Learning Objectives (ILOs):

1. Familiarize the students about the importance of materials science in modern civilization.
2. Students will classify metals, ceramics, polymers, and electronic materials in context of atomic structure and interatomic bonding.
3. The course will build up the students' ability to make the relation among processing, structure, and physical properties of materials.

Course Learning Outcomes (CLOs):
After successfully completing the course, the students should be able to:

1. Classify the materials.
2. Understand the basic properties that characterize the behavior of materials.
3. The course will build up the students' ability to make the relation among processing, structure, and physical properties of materials.

Course contents:

1. **Understanding of Materials:** Basic concepts of materials science and engineering; Development of materials; Uses of materials; Advanced materials; Modern materials' needs; Scope and applications of materials science and engineering.
2. **Types of Solid Materials:** Metal, polymer, ceramics, composites, semiconductor, crystalline and amorphous solids; Superconductor; Characteristics of materials.
3. **Properties of Materials:** Electrical and electronic properties; Thermal and thermoelectric properties; Mechanical Properties; Optical properties; Magnetic properties.
4. **Solidification of Materials:** Introduction; Nucleation and growth of crystal; Homogeneous and heterogeneous nucleation; Types of solid solution; Ordered and disordered solid solution; Grain and grain boundaries; Effect of cooling rate on grain size and mechanical properties.
5. **Diffusion in Solids:** Diffusion mechanisms; Steady-state and non-steady-state diffusions; Diffusion in porous media; Physical origin and rate equations; Transient diffusion; Factors that influence diffusion; Other diffusion path.
6. **Materials Selection and Design:** Introduction; Importance of materials selection; Factors affecting the selection of materials; Relation of materials selection to design; Product analysis; Activities of product development.

Recommended books:

- William D. Callister : Materials Science & Engineering- An Introduction
- William F. Smith : Foundation of Materials Science and Engineering
- L.H. Van Vlack : Elements of Materials Science and Engineering
- R.B. Gupta : Materials Science
- K.J. Pascoe : Properties of Engineering Materials

Course Code	Course Title	Credits	Marks	Contact Hours/Week
MSE1121	Crystallography-1	2	50	2

Course Types: Theoretical [x] Sessional/Practical [ ] Project/Training [ ] Viva-voce [ ]

Teaching-learning strategies: Lectures/Exercises/Quizzes/Class Tests/Attendance

Assessment Strategies: Attendance (10%); Quizzes/Class Test (20%); Examination (70%)

Course Description: The objective of this course is to present the basic concepts needed to understand the crystal structure of materials. The chapter 1 is designed including different types of primary and secondary bonds that are exist in solid materials. A thorough description of a crystal is assigned in topics of lattice, basis, unit cell, crystal system, atomic packing factor, lattice plane, lattice directions, and crystalline structure of metal, ceramics and polymer.

Intended Learning Objectives (ILOs):

1. Describe different types of bonds that are exists in solid materials.
2. Define lattice, crystal system, lattice plane and direction.
3. Explain different types of defects in crystals.

Course Learning Outcomes (CLOs):
After successfully completing the course, the students should be able to:

1. Calculate packing factor, density, interplanar spacing, using the geometrical parameters of crystal system.
2. Clarify which type of defect has severe impact on material properties.

Course contents:

1. **Crystalline Materials:** Crystalline and non-crystalline materials; Types of crystalline solids: ionic, covalent, molecular and metallic crystals; Cohesive energy of ionic crystals; Lattice energy; Born-Haber cycle; Isomorphism, polymorphism, enantiotropy and monotropy.
2. **Crystals and Crystal Structures:** Nature of crystalline states; Faces, edges and interfacial angle; Space lattice; Unit cells and patterns; Periodicity in crystals; Atomic packing: hcp and ccp structures; Construction of crystals: closed packed hexagonal and square layers of atoms, body-centred cubic crystal, and some simple ionic and covalent structures; Selected crystal structures: Pure metals, diamond and graphite, co-ordination in ionic crystals, AB-type compounds, silica, alumina, complex oxides, silicates; Crystallinity in polymers.
3. **Crystal Planes and Directions:** Arrangements of ions in crystals; Lattice planes; Indexing lattice directions and lattice planes; Miller indices and zone axis symbols; Lattice planes in cubic crystals: lattice plane spacing, inter-planar distance, ratio of lattice spacing; Miller indices and Laue indices; Zones, zone axes and the zone law; Addition rule; Indexing in the trigonal and hexagonal systems; Transforming miller indices and zone axis symbols.
4. **Crystal Imperfections:** Effects of imperfection on crystal; Causes of imperfections in crystals; Types of crystal imperfections/defects: Point defect, line defect, plane defects, volume defect.

Recommended books:

- F. Donald Bloss : Crystallography and Crystal Chemistry
- C. Hammond : Introduction to Crystallography
- C. Kittle : Introduction to Solid State Physics
- Uma Mukherji : Engineering Physics

Course Code	Course Title	Credits	Marks	Contact Hours/Week
MSE1122	Crystallography-I Sessional	1	25	3

Course Types: Theoretical [ ] Sessional/Practical [x] Project/Training [ ] Viva-voce [ ]

Teaching-learning strategies: Lectures/Exercises/Quizzes/Class Tests/Attendance

Assessment Strategies: Attendance (10%); Quizzes, Viva-voce and Continuous Assessment (30%); Practical Examination/ Design work/Report (60%)

Course Description: This course designed to develop strong fundamentals of crystal structure. Students will be able to identify different crystal system and calculate the density of the crystal. They will gain the knowledge to find out lattice parameter and defect phase from the XRD pattern of a particular material.

Intended Learning Objectives (ILOs):

1. Be familiar with unit cell, primitive cell and different crystal structures and systems.
2. Familiar with various types of defects in unit cell.

Course Learning Outcomes (CLOs):
After successfully completing the course, the students should be able to:

1. Visualize the unit cell of different crystal structures.
2. Visualize defects in unit cell of various crystal structures.

Course contents:

1. Concept of Unit and dimensions.
2. Familiar with different crystal systems.
3. Identification of different crystal structures.
4. Calculation of compactness of different structures.
5. Atomic Stacking and formation of common crystal structures.
6. Formation of voids.
7. Miller Indices of direction and plane.

Recommended books:

- C. Kittle : Introduction to Solid State Physics

Course Code	Course Title	Credits	Marks	Contact Hours/Week
MATH1111	Matrices, Vector Analysis and Co-ordinate Geometry	3	75	3

Course Types: Theoretical [x] Sessional/Practical [ ] Project/Training [ ] Viva-voce [ ]

Teaching-learning strategies: Lectures/Exercises/Quizzes/Class Tests/Attendance

Assessment Strategies: Attendance (10%); Quizzes/Class Test (20%); Examination (70%)

Course Description: This course is introducing to familiarize the students with linear algebra, trigonometry and vector analysis. This course provides the students a natural aid to the understanding of some physical concept in solving engineering problem.

Intended Learning Objectives (ILOs):

1. Introduce the fundamentals of algebra to solve mathematical equations.
2. Familiarize the students to understand trigonometric functions and calculating their value.
3. Present the fundamental concepts of vector and to develop student understanding and skills in the topic necessary for its applications to science and engineering.

Course Learning Outcomes (CLOs):
After successfully completing the course, the students should be able to:

1. Understand sets, relation and function to solve the systems of linear equation.
2. Use geometric functions of complex argument to model a variety of real-world problem-solving applications.
3. Perform basic matrix and vector operations both graphically and algebraically-addition, subtraction and scalar multiplication.

Course contents:

1. **Algebra:** Algebra of sets; De Morgan's rule, relation & function; De Moivre's theorem; Theorem, and relation between roots and coefficients; Solution of cubic equations; Determinants: Properties and Cramer's rule.
2. **Matrices:** Algebra of matrices; Evaluation of inverse, adjoint and rank of matrices with properties; Echelon, canonical and normal forms; Solution of system of linear equations by Gauss elimination and matrix method; Eigenvalues and eigenvectors; Caley-Hamilton theorem; Similar matrices and diagonalization.
3. **Co-ordinate Geometry of three Dimensions:** Cartesian, Cylindrical and spherical polar Co-ordinates; Transformation coordinates; Projections; Direction cosines and direction ratios; Angle between two lines; Plane: Different form of the equations of a plane, Plane through points and lines, Angle between two planes, Distance of a point from a plane, Volume of tetrahedron; Derivation of equations of straight lines; Line on a plane; Coplanar lines; Distance of a point from a line; Shortest distance; Equations of sphere and its tangent plane; Circles on spheres; Orthogonality of spheres.
4. **Vector Analysis:** Vector Addition, Multiplication, Scalar and vector products & Differentiation; Gradient of scalar functions; Divergence and curl of vector functions with their physical significance; Vector line, surface and volume integrations; Green's theorem; Stoke's theorem; Divergence Theorem and Gauss theorem; Applications of vector calculus to engineering problems.

Recommended books:

- H.S. Hall & S.R. Knight : Higher Algebra
- B.C. Das & B.N. Mukherjee : Higher Trigonometry
- M.R. Spiegel : Vector Analysis

• Barnside & Panton : Theory of Equations
• Rahman & Bhattacharjee : Coordinate Geometry with Vector Analysis
• Robert J. T. Bell : An Elementary Treatise on Coordinate Geometry of Three Dimensions

Course Code	Course Title	Credits	Marks	Contact Hours/Week
PHY1111	Electricity and Magnetism	3	75	3

Course Types: Theoretical [x] Sessional/Practical [ ] Project/Training [ ] Viva-voce [ ]

Teaching-learning strategies: Lectures/Exercises/Quizzes/Class Tests/Attendance

Assessment Strategies: Attendance (10%); Quizzes/Class Test (20%); Examination (70%)

Course Description: It is one of the fundamental or core course of physics those who want to acquire fundamental electricity and magnetism knowledge and to link them up with their studies in fields like engineering, chemistry and mathematics. In this course fundamental laws of electricity, thermoelectricity and magnetism are discussed to know their applications in aspect of modern physics. Problems solving and analytical skills of students will have increased by studying this course. From this course the students will gain primary knowledge on electrical AC and DC circuits of parallel and series combinations as well.

Intended Learning Objectives (ILOs):

1. To deliver knowledge on the fundamental laws of physics based on electrostatics.
2. To deliver idea about parallel plate capacitors using dielectrics and Gauss's law.
3. To give knowledge on electric current based on theory of electron.
4. To provide overall concept on electromagnetic induction using Faraday's and Ampere's law, traditional and functional ceramics, types and structural composition of ceramics.
5. To deliver knowledge on principles of thermoelectricity; DC and AC circuits of current in series parallel connections.
6. To provide techniques and knowledge how to analyze and solve physical problems with aids of mathematics.

Course Learning Outcomes (CLOs):
After successfully completing the course, the students should be able to:

1. Know the theory of capacitance, and its application using dielectrics and Gauss law and the electron theory of conductors.
2. Know the use of Coulomb's law and Gauss' law for the electrostatic force and the relationship between electrostatic field and electrostatic potential.
3. (a) Know the use of the Lorentz force law for the magnetic force and the use of Ampere's law to calculate magnetic fields. (b) Know the use of Faraday's law in induction problems and laws of thermoelectricity. (c) Know the basic laws that underlie the properties of electric circuit elements.

Course contents:

1. **Electrostatics:** Electric field; Potential difference and electric potential; Electric field and potential due to various charge distributions; Electric dipole; Electric field due to a dipole; Local electric field of an atom; Electric dipole on external electric field; Gauss's Law and its applications.
2. **Capacitance and Dielectrics:** Parallel plate capacitors; Susceptibility, permeability, and dielectric constant; Theory of dielectric constant, Gauss's theorem in dielectrics; Capacitors with dielectrics; Energy stored in an electric field.
3. **Thermoelectricity:** Thermal e.m.f; Seebeck, Peltier and Thomson Effects; Laws of addition of thermal e.m.f.; Thermoelectric power.

4. **Electric Current and Resistance:** Current and current density; Resistance and Ohm's law; The resistivity of different conductors; Electrical energy and power; Electron theory of conductivity: Conductor, semiconductors and insulators; Kirchhoff's laws; Superconductivity.
5. **Electromagnetic Induction:** Faraday's experiment; Faraday's law; Ampere's law; Motional e.m.f.; Self and mutual inductance galvanometers-moving coil; Ballistic and deadbeat types.
6. **A.C. Theory and Frequency Domain Analysis:** General AC theory, AC power, average and RMS value of AC voltage and current, resonance phenomena in circuits, Q-value and bandwidth.

Recommended books:

- Acharyya : Electricity and Magnetism
- Admas & Page : Principles of Electricity
- Emran et al. : Text Book of Magnetism and Electricity
- Halliday & Resnick : Physics (Part-I & II)
- Kip : Fundamentals of Electricity and Magnetism
- Huq et al. : Concept of Electricity and Magnetism

Course Code	**Course Title**	**Credits**	**Marks**	**Contact Hours/Week**
PHY1112	Electricity and Magnetism Sessional	1	25	3

Course Types: Theoretical [ ] Sessional/Practical [x] Project/Training [ ] Viva-voce [ ]

Teaching-learning strategies: Lectures/Exercises/Quizzes/Class Tests/Attendance

Assessment Strategies: Attendance (10%); Quizzes, Viva-voce and Continuous Assessment (30%); Practical Examination/ Design work/ Report (60%)

Course contents:

1. To study the design and construction of a capacitor.
2. To study the capacitive properties of dielectric materials.
3. To study the variation of reactance of R-L series circuit with frequency.
4. To study the variation of reactance of R-C series circuit with frequency.
5. To stud the design, construction and frequency response of a RLC series resonance circuit.
6. To study Faraday's law of electromagnetic induction.
7. To study the design and construction of a galvanometer.
8. To study the variation of Thermo-Emf of a Thermocouple with Difference of Temperature of its Two Junctions.

Course Code	Course Title	Credits	Marks	Contact Hours/Week
CHEM1111	Organic Chemistry	2	50	2

Course Types: Theoretical [x] Sessional/Practical [ ] Project/Training [ ] Viva-voce [ ]

Teaching-learning strategies: Lectures/Exercises/Quizzes/Class Tests/Attendance

Assessment Strategies: Attendance (10%); Quizzes/Class Test (20%); Examination (70%)

Course Description: This course is an introductory course in organic chemistry and intends to transmit basic knowledge of organic chemistry, which is required for the successful completion of other undergraduate courses of general background and/or specialization, such as materials chemistry, polymer chemistry/polymer materials and biomaterials.

Intended Learning Objectives (ILOs):

1. The students should able to develop basic skills for the multi-step synthesis of organic compounds and mechanism for a chemical reaction.
2. This course will help to understand the structure and reactivity of organic molecules, with examples illustrating the role of organic chemistry in industry.
3. They can apply chemical principles in the laboratory setting.

Course Learning Outcomes (CLOs):
After successfully completing the course, the students should be able to:

1. Gather and recall the fundamental principles of organic chemistry that include purification, elemental analysis and bonding.
2. Extend knowledge on nomenclature, preparation, properties, types of reactions and important uses of aliphatic hydrocarbons.
3. Understand the basic concept of biochemistry and stereochemistry of organic molecules.

Course contents:

1. **General Concept:** Purification of organic compounds; Detection of elements in organic compounds; Bonding: Covalent bond (σ and π) formation in organic compounds; Molecular shape; Delocalization and resonance.
2. **Aliphatic:** Nomenclature; Preparation, properties, types of reactions and important uses of aliphatic hydrocarbons; Mechanism of reaction and their homologous: Alkyl halides, Grignard reagents, Alcohols, Aldehydes ethers and ketones, Carboxylic acids and their derivatives, Alkyl amines.
3. **Alicyclic Compounds:** Nomenclature; Preparation and reactions of alicyclic compound: Cyclopropane, Cyclobutane, Cyclohexane and their derivatives; Ring formation and stability; Angle strain; Baeyer strain theory and its weakness; Sachse-Mohr modifications.
4. **Aromatic compounds:** Structure of benzene (Kekule & resonance structure); Aromaticity; Mechanism of electrophilic and nucleophilic substitution; Orientation and resonance; Effects of the substituted groups in the mono-substituted benzene rings (activation & deactivation, orientation).
5. **Biochemistry:** Definition, characteristics classification, structures, and reactions of amino acids (isoelectric points); Proteins and carbohydrates.
6. **Stereochemistry:** Stereoisomers; Asymmetric carbon atoms and optically active compounds; Optical and molecular rotation; R and S configurations; Optical and geometrical isomerisms of simple organic compounds.

Recommended books:

- R.T. Morrison & R.N. Boyd : Organic Chemistry
- I. L. Finar : Organic Chemistry
- Bhal & Bhal : Organic Chemistry Vol. I & II
- E.L. Eliel : Stereochemistry of Carbon Compounds
- P. Sykes : A Guide to Mechanism in Organic Chemistry
- K. Barsal : Organic Reaction Mechanism
- E.S. Gouldganic : Organic Reaction Mechanism

Course Code	Course Title	Credits	Marks	Contact Hours/Week
CHEM1112	Organic Chemistry Sessional	1	25	3

Course Types: Theoretical [ ] Sessional/Practical [x] Project/Training [ ] Viva-voce [ ]

Teaching-learning strategies: Lectures/Exercises/Quizzes/Class Tests/Attendance

Assessment Strategies: Attendance (10%); Quizzes, Viva-voce and Continuous Assessment (30%); Practical Examination/ Design work/ Report (60%)

Course Description: The purpose of the undergraduate general chemistry program is to provide the key knowledge base and laboratory resources to prepare students for careers as professionals in the field of material science.

Intended Learning Objectives (ILOs):

1. The planning and implementation of advanced organic reactions.
2. Basic chemistry techniques, such as how to calculate percent yields, how to use measuring devices properly.
3. Detection of elements in organic compounds.
4. Identification of functional groups organic compounds.
5. Determination and identification of different metal ions.

Course Learning Outcomes (CLOs):
After successfully completing the course, the students should be able to:

1. Identify metal and functional groups in organic compounds.
2. Purify and characterize of organic compounds.

Course contents:

1. Detection of elements in organic compounds.
2. Identification of functional groups organic compounds.
3. Identification of unknown organic compounds by their physical constants such as melting point and boiling point.
4. Synthesis of different organic compounds.
5. Separation, purification and characterization of organic compounds.
6. Analysis of water and some industrial products.

Course Code	Course Title	Credits	Marks	Contact Hours/Week
ENG1111	Technical and Communicative English	2	50	2

Course Types: Theoretical [x] Sessional/Practical [ ] Project/Training [ ] Viva-voce [ ]

Teaching-learning strategies: Lectures/Exercises/Quizzes/Class Tests/Attendance

Assessment Strategies: Attendance (10%); Quizzes/Class Test (20%); Examination (70%)

Course Description: Technical and communicative English is a part of the regular course and taught in the first year of the engineering degree. This course introduces grammatical principle, speaking, reading and writing section to develop communicative skill of the students.

Intended Learning Objectives (ILOs):

1. To develop and integrate the use of four language skills; listening, speaking, reading and writing abilities of the students.

Course Learning Outcomes (CLOs):
After successfully completing the course, the students should be able to:

1. Use English effectively for academic purposes across the curriculum.
2. Communicate effectively and appropriately in real life situations.

Course contents:

1. **Grammatical Principle:** Sentences; Tenses; Clauses; Connectors; Understanding long sentences; Conditional sentences; Transformation of sentences; Phrases: Noun phrases, adverbial phrases, prepositional phrases, verb phrases; Modifier; Head word; Appositives; Determiners; Making words with adding suffix and prefix; Phrase and idioms; Modal verbs; Gerund; Infinitive; Bear infinitives; Causative verbs; Words of negation.
2. **Spoken English:** Greetings; Dialogues; Responding to a particular situation; Extempore speech.
3. **Reading:** Comprehension of technical & non-technical materials-skimming, scanning, inferring & responding to context.
4. **Technical Report and Presentation:** Paragraph & composition writing on scientific & other themes; Report writing; Research paper writing; Library references.
5. **Professional communication:** Business letter; Job application; Memos; Quotations; Tender notice.

Recommended books:

- J. Thomson & A.V. Martinet : A Practical English Grammar
- John M. Lennon : Technical Writing
- A. Ashley : Oxford Handbook of Commercial Correspondence
- J. Swales : Writing Scientific English
- Robert J. Dixson : Complete Course in English
- Rajendra Pal & J.S. Korlahalli : Essentials of Business Communications

Course Code	Course Title	Credits	Marks	Contact Hours/Week
MSE1132	Engineering Drawing	1	25	3

Course Types: Theoretical [ ] Sessional/Practical [x] Project/Training [ ] Viva-voce [ ]

Teaching-learning strategies: Lectures/Exercises/Quizzes/Class Tests/Attendance

Assessment Strategies: Attendance (10%); Quizzes, Viva-voce and Continuous Assessment (30%); Practical Examination/ Design work/ Report (60%)

Course Description: The central issue of this course is to help students to visualize the basics of engineering design and drawing. It is also expected to improve hands-on experiences towards their practical life.